PHASE - 1

1. Problem Statement

In educational institutions, multiple academic, technical, cultural, and extracurricular events are conducted throughout the year. Students often face difficulties in discovering relevant events, tracking registration deadlines, receiving timely updates, monitoring event status, and accessing participation certificates after event completion.

Currently, event-related information is frequently distributed across multiple channels such as notice boards, emails, messaging groups, and social media platforms. This fragmented approach can lead to missed announcements, delayed registrations, lack of awareness regarding ongoing and upcoming events, and challenges in obtaining certificates. Additionally, event organizers, faculty coordinators, and student coordinators face difficulties in managing registrations, participant records, attendance tracking, event coordination, and certificate distribution efficiently.

The absence of a centralized event management system results in:

- Difficulty in discovering and tracking upcoming, ongoing, and completed events.
- Missed registration opportunities due to scattered announcements.
- Inefficient management of participant registrations and team-based events.
- Lack of a streamlined communication mechanism between organizers and students.
- Manual attendance tracking processes that are time-consuming and prone to errors.
- Delays and administrative overhead in generating and distributing certificates.
- Limited visibility into event participation statistics and analytics.
- Lack of a structured event lifecycle and approval process, leading to inconsistencies in event administration.
- Difficulty in monitoring event activities, changes, and operational accountability due to the absence of audit and tracking mechanisms.

To address these challenges, there is a need for a centralized digital platform that enables students to discover events, register individually or as teams, receive real-time notifications, track participation, mark attendance, and download certificates seamlessly. The platform should also provide administrators, faculty coordinators, and student organizers with comprehensive tools for event creation, approval management, registration monitoring, waitlist handling, attendance tracking, certificate generation, notifications, analytics, event coordination, budget planning, and post-event feedback analysis.

The proposed solution, Avento, aims to streamline the complete event lifecycle by providing an integrated and user-friendly event management system for students, faculty coordinators, student organizers, and administrators. The platform seeks to improve accessibility, operational efficiency, governance, transparency, and engagement across institutional events

through centralized event management, automated workflows, attendance monitoring, certificate distribution, and analytical insights.

2. Proposed Solution

Avento is a centralized event management platform designed to streamline the planning, approval, registration, participation, attendance tracking, feedback collection, and certificate distribution processes within an educational institution.

The platform provides a unified system where students can discover upcoming, ongoing, and completed events, register for events individually or as teams, receive timely notifications, manage registrations, access waitlists when events reach capacity, and download certificates after successful participation. At the same time, administrators, faculty coordinators, and student organizers can efficiently manage event creation, registrations, attendance records, communications, budget planning, and certificate generation through dedicated dashboards.

To ensure authenticity and reduce manual effort, the system incorporates institutional email-based authentication with OTP verification, structured event lifecycle management, attendance tracking, automated event updates, and certificate generation using predefined templates. The platform also supports event approval workflows, waitlist management, participation analytics, post-event feedback analysis, budget planning, and activity monitoring to improve operational transparency and governance.

By consolidating all event-related activities into a single platform, Avento aims to improve student engagement, simplify administrative workflows, enhance coordination among organizers, support better event planning, and provide a seamless event management experience for all stakeholders within the institution.

Note : Why the Name Avento?

The name **Avento** is inspired by the word **Event** and represents a modern platform that brings together every stage of an event journey—from event creation and approval to registration, participation, attendance tracking, notifications, feedback collection, analytics, and certificate generation. The name reflects simplicity, innovation, and a seamless event management experience within a single ecosystem.

Avento symbolizes a centralized space where students, organizers, faculty coordinators, and administrators can collaborate efficiently, making institutional event management more organized, accessible, transparent, and engaging.



3. How Avento is Different from Existing Solutions

Most educational institutions rely on multiple disconnected methods such as notice boards, emails, WhatsApp groups, Google Forms, spreadsheets, and paper-based registration processes for event management. While these methods serve basic requirements, they often lead to fragmented communication, duplicate data entry, manual administrative effort, and reduced student participation.

Avento differentiates itself by providing a centralized and integrated platform that manages the complete event lifecycle within a single system.

Key Differentiators

3.1 Centralized Event Management

Unlike traditional approaches where event information is distributed across notice boards, emails, messaging groups, and registration forms, Avento provides a single platform for viewing upcoming, ongoing, and completed events.

3.2 Structured Event Lifecycle Management

Avento introduces a well-defined event lifecycle including Draft, Pending Approval, Published, Registration Open, Registration Closed, Ongoing, Completed, Archived, and

Cancelled states. This ensures better event governance and transparency compared to conventional event management approaches.

3.3 Secure Institutional Authentication

Students can register only using their official college email addresses with OTP verification, ensuring authenticity and preventing unauthorized access.

3.4 Role-Based Management Structure

Unlike conventional systems where a single administrator handles all operations, Avento supports multiple roles including Super Admin, Faculty Coordinator, Student Organizer, and Student, enabling better responsibility distribution and event coordination.

3.5 Individual and Team-Based Registration

Unlike conventional Google Forms and manual registration methods, Avento supports both individual and team registrations within the same platform, including team creation, leader-based registration, team size validation, and member notifications.

3.6 Automated Registration Control

Traditional registration systems often require organizers to manually monitor participant limits through spreadsheets. Avento automatically closes registrations once the event capacity is reached and supports waitlist management, allowing vacant seats to be filled when registered participants cancel.

3.7 Smart Attendance Tracking

Unlike paper attendance sheets, Avento provides a digital attendance tracking mechanism with session-based attendance management, reducing manual effort and improving accuracy.

3.8 QR Code-Based Attendance

Avento supports QR code-based attendance marking, allowing participants to quickly check in to event sessions by scanning a generated QR code. This minimizes manual verification, reduces attendance fraud, speeds up participant processing, and improves attendance accuracy.

3.9 Automated Certificate Generation

Unlike traditional certificate distribution methods that require manual preparation and sharing, Avento automatically generates certificates using predefined templates and provides them only to eligible participants based on attendance requirements.

3.10 Real-Time Notifications

Students receive instant updates regarding registrations, team registrations, event schedules, attendance, event changes, cancellations, and certificate availability through both application notifications and email alerts.

3.11 Dedicated Student Dashboard

Students can access all event-related information, registrations, attendance records, notifications, participation history, and certificates through a personalized dashboard instead of searching across multiple communication channels.

3.12 Approval-Based Event Publishing

Faculty Coordinators can create events, which are reviewed and approved by the Super Admin before publication. This prevents unauthorized, incomplete, or duplicate events from being published.

3.13 Administrative Analytics

Administrators gain access to event statistics, registration trends, attendance reports, participation analytics, department-wise participation metrics, event popularity rankings, feedback insights, and certificate distribution records, which are difficult to obtain using spreadsheets and manual records.

3.14 Activity Monitoring and Audit Trail

Avento maintains activity logs for key operations such as event creation, modifications, approvals, attendance actions, and certificate generation. This improves accountability and operational transparency.

3.15 End-to-End Event Lifecycle Management

Avento manages the complete event lifecycle, including event creation, approval, registration, team management, attendance tracking, QR-based attendance verification, notifications, analytics, audit logging, and certificate distribution within a single platform, eliminating the need for multiple disconnected tools.

Feature	Traditional Method (Forms / Sheets / Paper)	Avento
Event Discovery	Multiple Sources	Single Centralized Platform
Authentication	Open Registration / Manual Verification	College Email + OTP Verification

Role Management	Single Administrator	Super Admin, Faculty Coordinator, Student Organizer & Student Roles
Event Lifecycle	Basic Event Posting	Structured Lifecycle with Approval Workflow
Event Approval	Not Available	Super Admin Approval Process
Registration	Google Forms / Paper Forms	Integrated Registration System
Team Registration	Manual Collection of Member Details	Automated Team Registration & Validation
Participant Limit Control	Manual Monitoring	Automatic Registration Closure
Attendance Tracking	Paper Sheets / Manual Verification	QR-Based Digital Attendance
Multi-Session Attendance	Difficult to Manage	Session-Based Attendance Tracking
Notifications	WhatsApp / Email	In-App Notifications + Email Alerts
Event Updates	Manual Communication	Automated Notifications & Dashboard Updates
Certificate Distribution	Manual Preparation & Sharing	Automated Generation & Download
Certificate Eligibility	Manual Verification	Attendance-Based Eligibility Validation
Analytics	Limited Reports	Real-Time Analytics Dashboard
Participation Insights	Manual Calculation	Department, Team & Event Analytics
Audit Trail	Not Available	Activity Logging & Monitoring
Data Management	Multiple Files & Spreadsheets	Centralized Database
Overall Event Management	Multiple Disconnected Tools	End-to-End Integrated Platform
Registration Cancellation & Waitlist	Usually Not Available	Cancellation Support with Automatic Waitlist Promotion

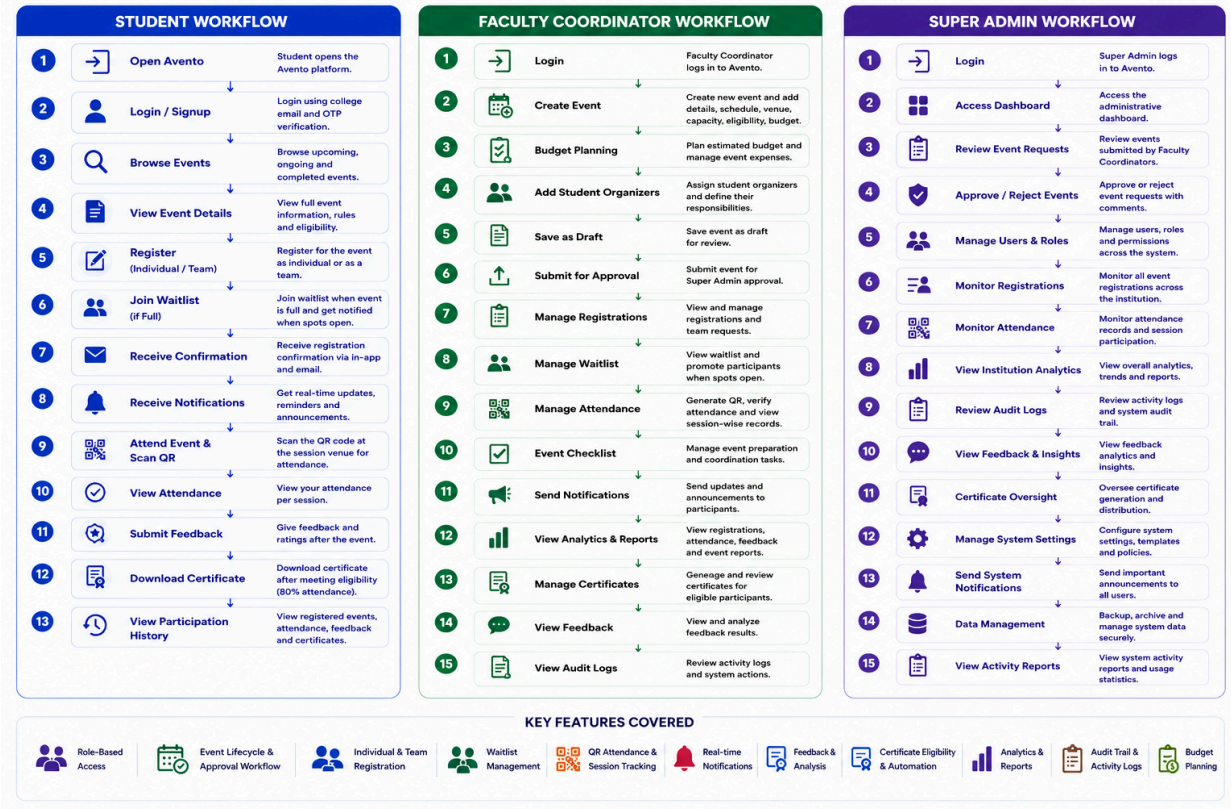
Feedback Collection	Manual Surveys / Google Forms	Integrated Post-Event Feedback & Analysis

4. Roles and Their Modules

Module	Super Admin	Faculty Coordinator	Student Organizer	Student
Authentication & Access	✓	✓	✓	✓
User & Role Management	✓	✗	✗	✗
Event Creation & Management	✓	✓	✗	✗
Event Approval	✓	✗	✗	✗
Event Discovery	✓	✓	✓	✓
Registration (Individual / Team)	✗	✗	✗	✓
Registration Monitoring	✓	✓	✓	✗
Team Management & Monitoring	✓	✓	✓	✓
Waitlist Management	✓	✓	✓	✓
Attendance & QR Verification	✓	✓	✓	✓
Notifications & Announcements	✓	✓	✓	✓
Certificates (Generate / Download)	✓	✓	✗	✓
Feedback & Analysis	✓	✓	✓	✓
Budget Planning	✗	✓	✗	✗
Analytics & Reports	✓	✓	✓	✗
Audit Trail & Activity Logs	✓	✓	✗	✗

Personal Dashboard & History	✗	✗	✓	✓
Event Coordination Checklist	✗	✓	✓	✗

5. Role Wise Flow Chart



6. Functional Requirements

ID	Requirement Area	Description
FR-01	User Registration & Verification	Allow users to register using institutional email addresses and verify accounts through OTP authentication.
FR-02	User Authentication	Enable secure login and logout functionality for authorized users.

FR-03	Role-Based Access Control	Restrict system functionalities based on user roles (Super Admin, Faculty Coordinator, Student Organizer, Student).
FR-04	User & Role Management	Allow Super Admin to manage users, roles, and permissions.
FR-05	Event Creation & Management	Allow authorized organizers to create, edit, configure, and manage events.
FR-06	Event Approval Workflow	Enable Super Admin to approve or reject events before publication.
FR-07	Event Lifecycle Management	Manage event states such as Draft, Published, Registration Open, Registration Closed, Ongoing, Completed, and Cancelled.
FR-08	Event Discovery & Information Display	Allow users to browse events and view event details including date, venue, description, eligibility, and participant limits.
FR-09	Event Registration Management	Support individual and team registrations, participant validation, team notifications, and participant limit enforcement.
FR-10	Registration Monitoring	Allow organizers to monitor registrations and participant information.
FR-11	Waitlist Management	Allow eligible users to join waitlists when events reach capacity and automatically promote waitlisted participants when vacancies arise.
FR-12	Attendance Management	Generate QR codes, verify attendance through QR scanning, and maintain attendance records.
FR-13	Certificate Management	Manage certificate templates, generate certificates automatically for eligible participants, and allow certificate downloads.
FR-14	Notification Management	Send reminders, announcements, updates, venue changes, registration updates, waitlist promotions, and cancellation notifications.
FR-15	Dashboard Management	Provide student and organizer dashboards with relevant event, attendance, registration, and certificate information.

FR-16	Feedback Management	Allow participants to submit post-event feedback and enable organizers to analyze feedback results.
FR-17	Budget Planning Management	Allow Faculty Coordinators to create and manage basic event budget plans and estimated expenses.
FR-18	Analytics & Reporting	Generate reports and analytics related to registrations, attendance, participation, event popularity, feedback, and certificates.
FR-19	Audit Trail Management	Record significant system activities including event approvals, modifications, attendance actions, and certificate generation.
FR-20	Checklist Management	Allow Faculty Coordinators and Student Organizers to manage event preparation and coordination tasks.
FR-21	Data Storage & Security	Securely store and protect user, event, registration, attendance, certificate, and audit data.

Note: LLM Models and Prompts

Currently, Avento does not utilize any Large Language Models (LLMs). The platform focuses on core functionalities such as event management, registrations, attendance tracking, notifications, analytics, and certificate generation.

As a future enhancement, AI-powered features such as event recommendations, intelligent search, FAQ chatbots, and virtual assistance may be integrated to further improve the user experience.